



Assisting Drivers with Ambient Take-Over Requests in Highly Automated Driving

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ABSTRACT

Take-over situations in highly automated driving occur when drivers have to take over vehicle control due to automation shortcomings. Due to high visual processing demand of the driving task and time limitation of a take-over maneuver, appropriate user interface designs for take-over requests (TOR) are needed. In this paper, we propose applying ambient TORs, which address the peripheral vision of a driver. Conducting an experiment in a driving simulator, we tested a) ambient displays as TORs, b) whether contextual information could be conveyed through ambient TORs, and c) if the presentation pattern (static, moving) of the contextual TORs has an effect on take-over behavior. Results showed that conveying contextual information through ambient displays led to shorter reaction times and longer times to collision without increasing the workload. The presentation pattern however, did not have an effect on take-over performance.

ACM Classification Keywords

H.5.m. Information Interfaces and Presentation (e.g. HCI): Miscellaneous

Author Keywords

Ambient light display; attention; automated driving; driver; hand over; take-over request.

INTRODUCTION

With the rise of advanced driving assistant systems (ADAS), highly automated driving has become more probable. In the past years automotive industry has introduced automated vehicles as the next disruptive innovation in the market. However, according to NHTSA [1], the transition from manual to automated driving happens gradually and in different levels. They predict that, the next generation of vehicles will be at level 3 of automation which is "Limited Self Driving Automation". This means that the driver does not need to monitor the roadway



Figure 1: Exploring peripheral visual cues as take-over requests

constantly but still be available to take over control in occasional cases of hazard with sufficiently comfortable transition times. These situations are known as *take-over* situations [20].

Highly automated driving can increase drivers' safety due to reduction of human failures, and comfort for relieving them from driving task [5]. This can change the role of the drivers and encourage them to engage themselves in more secondary tasks such as reading, surfing the internet, etc. [7]. Consequently, the in-vehicle environment becomes a multi-tasked context where drivers switch between driving and their secondary tasks.

In take-over situations, drivers have to switch from their secondary task to the driving task. During this switch, they have to shift their attention from one task to the other, which requires perceiving the state of the driving environment, make decisions, and act accordingly. Being engaged with secondary tasks can cause drivers to be vulnerable to delays or errors when getting back to the driving task due to not having a chance for attending to it and not being in the loop [16]. This can lead to hazardous situations if the driver has to take over vehicle's control due to automation shortcomings. Therefore, appropriate user interface designs for *Take-over Requests (TOR)* are required to ensure smooth transition from secondary to driving task.

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Researchers have run experiments on design, effectiveness and timings of take-over requests [20, 6, 18, 15]. However, to our knowledge, the existing work has investigated the effectiveness of TORs in isolation and without considering the multi-task context of driving and high amount of visual information that drivers have to process. A futuristic in-vehicle environment that relies heavily on control of automation raises a vital question: How can we support a driver's ability to seamlessly switch from engaging with a non-vehicle-handling task to monitor and/or resume the diverse complex maneuvers that constitute effective vehicle handling?

One aspect that addresses this question is the scheduling of visual information processing. In a dynamic and multi-task environment like driving, it is vital to define methodologies for effective communication of information by visual computing i.e. the process of taking complex data and presenting it in visual format that is easy to understand.

In this work, we follow an approach to support take-over situations using ambient visual displays. Our work in this paper yields three main research contributions:

- first, we show that while having audio cues to prime users with the urgency of take-over situation, locating the visual cue in the periphery (namely, a peripheral light display) can reduce mental workload and assist safe maneuvers
- second, our designed light display can convey contextual information to assist steering at take-over situations
- third, using different light patterns for presenting contextual information can have an effect on driving behavior.

The rest of this paper is structured as follows: first, we provide the background and survey of relevant related work, followed by our approach for designing peripheral light display TORs. Then, we present the details of our experiment design, followed by our results and discussion. Finally, we conclude and provide an outlook for future work.

RELATED WORK

Designing TORs has been the topic of research in recent years. Gold et al. [6] have run an experiment on timings of presentation of the TORs. They showed that shorter take-over times result in faster but worse reactions. Koo et al. have explored verbal message contents for situations when the automation takes over control from the driver. Their results showed that "how messages" (messages describing actions of automation) were less preferred and led to worse driving performance in comparison to "why messages" (messages describing reasons for automation actions) [9]. Naujoks et al. have run studies on designing TORs in auditory and visual modalities. They used a symbol showing hands on steering wheel in the information cluster display for the visual message. They found out that audio-visual messages result in shorter reaction time and better lateral control of the vehicle than only visual messages [18]. Baldwin & More studied use of different words to convey different levels of urgency to drivers, they found out that the word "danger" gain higher rating of urgency than the words "warning" and "caution" [2]. Based on this work, Politis et al. ran an experiment of multi-modal language based warnings for

take-over situation [20]. They investigated all combinations of audio, visual, and tactile warnings, and measured perceived urgency, annoyance and the effectiveness of the warnings. Their results indicated that increased number of modalities enhance perceived urgency and annoyance, higher urgency warnings result in shorter transition time, and unimodal visual warnings result in poor driving behavior.

All the studies above have used visual cues which are presented in forms of icons/symbols on the information cluster or head-up displays. These messages require focal visual attention which is used for fine detail and pattern recognition. This is the same visual channel required for performing the visual tracking task while driving. Considering the time limitation in take-over situation, according to Wickens' multiple resource theory, this can create a bottleneck in the time sharing of visual processing tasks [24]. Therefore, in this work, we propose using peripheral light displays for conveying take over request to drivers. According to Leibowitz et al. [12], peripheral displays address peripheral vision, which demands separate resources as focal vision in terms of supporting efficient time sharing.

Peripheral displays, convey information without demanding much attention or cognitive effort and are not involved with users' primary task [14, 23]. Due to these characteristics, and high visual workload in driving context, researchers have been studying the application of peripheral displays in cars. Laquai et al. [11] used ambient light displays to indicate future deceleration. Langlois et al. [10] used peripheral light reflected on the windshield to match problems signaled by advance driving assistant systems; this display indicated information such as distance alerts and lateral deviation. Loecken et al. [13] applied ambient light for a lane change assistant indicating the distance to other cars and the appropriate time to overtake. Meschtscherjakov et al. [17] used ambient light displays to assist drivers to maintain a predefined speed. These works show that, peripheral light displays can be used as a means to convey information in a non-obtrusive fashion to drivers.

DESIGNING AMBIENT TAKE-OVER REQUEST

As mentioned earlier, TORs perform their role in two stages: first, they shift the drivers' attention away from the secondary task, and second, they provide them with information to perform appropriate actions. Based on NHTSA definition of level 3 of automation, we assume take-over situation as situations where the driver has to take back the control of the vehicle due to automation shortcomings in sufficiently comfortable transition times [1]. Therefore, in the first stage, TORs require to be salient enough to attract drivers' attention. According to Politis et al. and Baldwin et al. [21, 2], the multimodal cues convey higher level of urgency than unimodal ones. This perception of urgency however, does not always improve performance. Work by Baldwin & May shows that collision avoidance was best with warnings with not too high or too low level of perceived urgency [3].

Based on these results, we decided to design our TOR as a bimodal display to add one more modality to our ambient visual TOR to communicate the urgency of the situation without stressing the drivers. Therefore, we chose to have an auditory

cue presented simultaneously with the light cue. Auditory warnings are recognised faster and are rated less annoying [21]. In line with [18], we used a sine wave tone with 1000HZ frequency and duration of 250 ms.

The light cues were presented via an illuminated LED strip located behind the steering wheel. Our first goal in this work was to investigate whether ambient light displays can be effective when used to prime a take-over situation. Therefore, we decided to light up the whole LED strip, simultaneously with the auditory cue to be an indication for the situation.

In a fully automated scenario, when drivers have sufficient trust in automation, it can be foreseen that they engage themselves more in secondary tasks. Therefore, a take-over maneuver not only requires disengagement from the secondary task, but also does demand taking them back to the loop. The second goal of this work was to pursue whether presenting contextual information as TORs to drivers, affects their performance. We aimed to investigate whether contextual information can be encoded and conveyed through light cues, and assist drivers to get back to the loop and be aware of the situation. Thus, we designed light cues which not only inform drivers about an upcoming take-over situation, but also do hint them about the steering direction when there's an obstacle on the way.

Our third objective, which is built on top of the second one, was to investigate whether the presentation pattern of the light cues have an effect on drivers' performance. We designed two different patterns for shifting the visual attention of the driver by hinting the direction of the obstacle: 1) static light 2) moving light. According to Neville & Lawson [19], moving stimuli has significantly larger effect for shifting peripheral attention than stationary stimuli. Thus, our assumption is that, information presented by moving light is perceived better than static light.

METHOD

Equipment

Take over requests were examined in a driving simulator to ensure repeatable outcomes for comparison, and to minimize the risk to drivers engaged in a distracting task on-road. A fixed-based right-hand traffic driving simulator with a field of vision of 150° was used. The simulation was created with SILAB¹. Auditory cues were also played simultaneously from speakers built in the driving simulator, located behind the driver on both sides.

As mentioned before, peripheral light displays were chosen to present visual cues. In this study, an Adafruit NeoPixel Digital RGB LED strip with a resolution of 144 LEDs per meter was used. To reduce the intensity of the light display, the LED strip was placed in a matte white acrylic LED profile. The frame was located on the dashboard of the driving simulator behind the steering wheel, 65 degrees from fixation on the tablet pc presenting the 1-back task, which was on the drivers' laps.

To detect the eye gaze of the participants during the experiment, they were asked to wear Dikablis Glasses by Ergoneers

². The eye-tracker was calibrated before each trial to ensure constant track of eye-gaze behavior. Two physical markers on the front panel and two virtual ones on the simulator displays were used. The calibration procedure took between 30 seconds to one minute for each trial. We used the standard eye-tracker software for calibration, video recording and analysis of participants' eye-gaze.

Participants

Twenty one (10 female, 11 male) students and research assistants ($M = 26.33$, $SD = 4.02$) took part in our experiment. They had between 1 to 19 years of driving experience. Participation in this study was voluntarily and participants were rewarded €15 for one and half hour.

Experiment Design and Measures

In our experiment, we chose a repeated measures within-subject design. This was done in order to ensure that all participants underwent all the light cue conditions. We had one independent variable which was the light condition. In all conditions we measured the following dependent variables:

- Reaction time (RT): the time between presentation of the TOR and first steering action,
- Time to collision (TTC) to obstacle: the time between the lane change maneuver and collision to the road block,
- Workload: perceived workload ratings by using NASA Raw Task Load Index (RTLX) [4]; 6 scales and overall task load index, and
- Eye-gaze behavior: number and duration of glances at the light display when the TORs were presented

We also collected qualitative feedback from the participants about the light displays and their influence on take-over task.

Procedure

Driving Task

The driving scenario was taking place in a highly automated car which was driving in the center lane of a 3-lane highway. The participants were asked to activate the automation in the beginning of each trial by pulling the cruise control lever. We asked the participants to start with an 1-back task on the tablet PC located on their laps having their hands on both sides of it, as soon as the automation was activated. They were encouraged to perform their best at the 1-back task and keep their eyes on it. In 30-40 second intervals a light and an audio cue was presented as TOR, informing them of a road block (a truck with road construction signs and alerts parked on the road) on either left or the right lane. The TORs were presented at 5 seconds TTC to the road block, in line with Gold et al. [6]. Due to this obstacle, the other cars on the road were changing their lanes, putting the drivers in the situation to either drive to the lane opposite to the block or brake. We asked our participant to change lane in all trials and just use the brakes in cases where they could not perform the lane-change maneuver. As soon as the drivers moved the steering wheel,

¹<https://wivw.de/en/silab>

²<http://www.ergoneers.com>



Figure 2: Spatial 1-back task: if the position of the lit up square is the same as the time before, the button "position" should be pressed.

the control of the vehicle was given back to them. After they have passed the road obstacle, they had to switch back to the center lane, activate the automation, and continue the 1-back task (Figure 1).

Secondary Task

The secondary task was spatial 1-back task presented in an android application³. The participant performed this task on a tablet PC which they laid on their laps. The task consisted of a 3×3 table. In one second intervals, one of the cells was lit up for duration of one second. When the lit up cell was at the same position as the previous one, they had to press a button labeled "position". This task was chosen since it requires focus of attention and increases working memory load. To have participants to be engaged in this task, we encouraged them to perform their best by showing them their records (Figure 2).

Conditions

Three test conditions were defined for each of our TOR designs:

- **Baseline:** The whole light display was turned on in red⁴ (in line with [20]), informing the driving of the upcoming take-over situation (Figure 3a)
- **Static Light:** The left or the right half of the light display was turned on in red informing the driving of the upcoming take-over situation and the location of road block (Figure 3b)
- **Moving Light:** The left or the right half of the light display was turned on in red with a moving pattern informing the driving of the upcoming take-over situation and the location of road block (Figure 3c). Since the LED strip had 144 LEDs and take over requests were presented at 5 seconds TTC, for the moving pattern 10 of the 72 LEDs (half of the display) were going on with a delay of 12 ms and iterating through the display.

Each of these conditions consisted of 30 trials (take-over situations). To avoid bias caused by learning, the order of conditions was counterbalanced between participants.

³<https://play.google.com/store/apps/details?id=cz.wie.p.nback>

⁴Red = RGB(255,0,0), brightness = 10 (0 being the dimmest and 255 the maximum brightness)

Pilot Study

Prior to our experiment, we ran a pilot study with 5 participants to examine the experiment setup. Our observations and qualitative feedback of the participant showed that having light direction in conditions 2 and 3, hinting the location of the obstacle was not found intuitive. All participants steered towards the direction that the light was cuing. This was despite the fact that the light was presented in red which is an indication for danger or hazard. Therefore, we redefined conditions 2 and 3 as: the left or the right half of the light display was turned on in red (with a moving pattern for moving condition) informing the drivers of the upcoming take-over situation and the direction to steer to (Figure 3). Based on qualitative results collected from this study, the location, brightness of the light display, and the position of the tablet PC were also optimized.

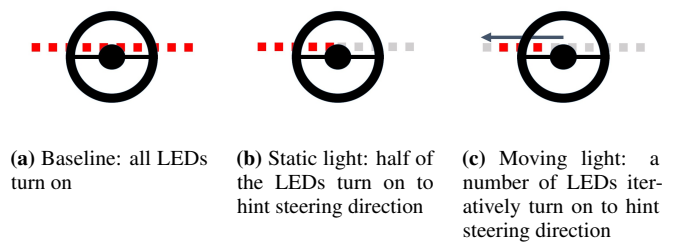


Figure 3: Light conditions: all images direct to the left side

Training Time

Each participant had between 3 to 5 minutes training time to get familiarized with both 1-back task on the tablet PC and driving and take-over task. They performed both tasks in parallel (exactly as defined in the experiment scenario) for 2 to 5 trials until they felt comfortable with the tasks.

Hypotheses

As mentioned earlier, we had three main objectives in this work: first, to investigate whether ambient light displays could be used as TORs; second, whether conveying contextual information in ambient light TORs can improve take-over maneuvers, third, whether the light pattern used for conveying contextual information can have an effect on take-over. Due to the conveyed contextual information in static and moving light conditions, we assumed that it simplifies decision making at take-over situations and thus will result in shorter reaction times and less risky maneuvers than baseline condition. As mentioned in design section, moving stimuli is proved to be perceived better than stationary ones. Therefore, we hypothesized that the reactions in moving light condition are faster and less risky than in static light condition. Similarly, for perceived overall task workload, we assumed lower ratings for static and moving conditions due to their assistant functionality; and lower ratings for moving condition in comparison with static, because of easier perception of movement. Following, is the list of hypotheses tested in our experiment:

- **H1:** the reaction time in static and moving light conditions will be less than baseline condition

- **H2:** the reaction time in moving light condition will be less than static condition
- **H3:** the time to collision to obstacle in static and moving light condition will be more than baseline condition
- **H4:** the time to collision to obstacle in moving light condition will be more than static condition
- **H5:** the overall NASA RTLX in baseline condition is higher than static and moving light condition
- **H6:** the overall NASA RTLX in static condition is higher than moving condition

RESULTS

Reaction Time and TTC

In all trials we measured the reaction time defined as the time between presentation of TOR and first steering action taken by participants. Figure 4 (left), shows the reaction times in milliseconds in the three light conditions. We also measured the time to collision to obstacle (TTC) defined as time between the lane change maneuver and the road block. This measure was used to observe the driving behavior of participants in different light conditions and how risky their maneuvers are. Figure 4 (right), shows the TTC means in seconds in all light conditions.

There was a main effect of cue-type for TTC ($F_{2,38} = 7.70, p = 0.002, \omega^2 = 0.246$) and RTs ($F_{2,38} = 7.46, p = 0.002, \omega^2 = 0.240$). Post-hoc Tukey HSD tests on both measures revealed that both static and moving cue conditions were significantly different from the baseline cue condition but not from each other. We performed a two-sided Bayesian hypothesis test (with a default Cauchy prior width of 0.707) to quantify the hypothesis that there is no performance difference between the static and moving cue conditions. The Bayes Factor (BF01) was 4.30 for TTC (which means that the null effect hypothesis is 4.3 (or 3.58) times more likely to be true than the alternative hypotheses) and 3.58 for RTs. This represents 'substantial' evidence that both types of cues are no different with regards to their performance benefits [8, 22].

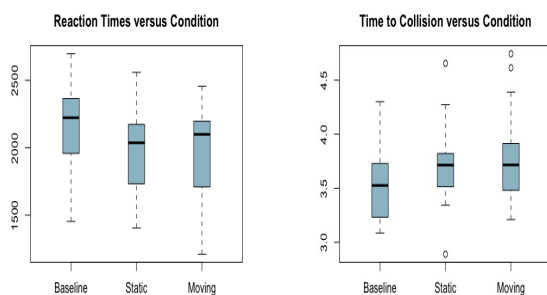


Figure 4: Reaction time means (left), and TTC means in three light conditions(right)

NASA Raw Task Load Index

After performing the tasks in each condition, we asked participants to fill in NASA RTLX questionnaire and rate perceived

workload of the task. The overall RTLX results showed required workload in performing the tasks in moving light condition ($M = 27.14, SD = 16.32$) was less than static light condition ($M = 28.57, SD = 16.03$), which was less than baseline condition ($M = 33.21, SD = 12.19$). Mauchy's test indicated that the assumption of sphericity was violated, $\chi^2(2) = 3, p = 0.223$. After using Huynh Feldt corrections ($\epsilon = 0.94$), the ANOVA test results showed no significant difference $F(1.89, 37.95) = 2.16, p = 0.132$. Figure 5 shows the means of all six RTLX scales in the three light conditions. We can see that the moving light, except for in frustration and temporal demand scales was rated the lowest (in performance, lower value indicates better performance). In conditions where the TOR light was cuing the drivers with the steering direction (static and moving), the temporal demand is the scale with highest value. This scale however, in the baseline condition is the second highest after mental demand. This can indicate that, cuing drivers with the steering direction can reduce mental workload for task switching and decision making.

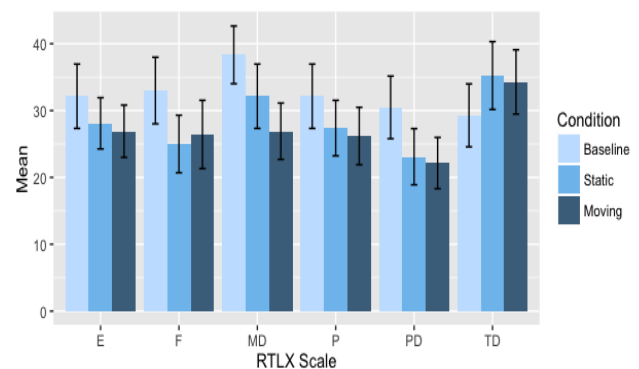


Figure 5: Nasa RTLX results for the three TOR light conditions. E: effort, F: frustration, MD: mental demand, P: Performance, PD: physical demand, TD: temporal demand

EYETRACKING ANALYSIS

In all three light conditions, we measured the number of glances on the light display when TOR was presented, we also measured duration of glances in these conditions. Figure 6 shows means and standard deviations of number of glances and duration of glances in all three light conditions.

We observed no main effect of cue type on duration of glances ($F_{2,38} = 2.235, p = 0.121, \omega^2 = 0.057$) and the number of glances, ($F_{2,38} = 3.088, p = 0.057, \omega^2 = 0.092$). We performed a JZS Bayesian t-tests in order to understand how the manipulated cues of static and moving compared to the baseline. In terms of number of glances, the static cue ($BF_{01}=0.21$) was more likely, than the moving cue ($BF_{01}=0.49$), to be different from the baseline. In addition, the mean duration of these glances were more likely to be different for the static cue ($BF_{01}=0.8$), than the moving cue ($BF_{01}=1.66$), to the baseline.

Using the labels provided by [22], we have 'substantial' evidence that static cues attract more glances than the baseline but

only 'anecdotal' evidence for moving cues. Furthermore, we have 'anecdotal' evidence that moving cues result in glances that have similar duration lengths as our baseline cues, and 'anecdotal' evidence that static cues result in longer glances.

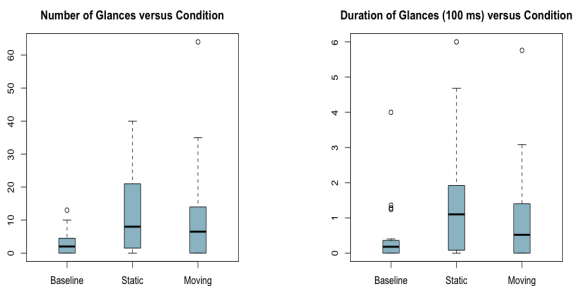


Figure 6: Means and standard deviations for number of glances (right) and duration of glances (left)

Qualitative Feedback

After the participants have performed the tasks under the three conditions, we interviewed them about how they find the light display, if they want to use it as TOR, and what problems did they see using it.

All participants said that they find the light displays useful and in general not annoying or stressing. They mentioned that having light in the periphery together with the auditory cue, attracts their attention faster to the handover task. 85% of the participants preferred conditions with contextual cuing (static and moving) more helpful than the baseline light. They mentioned "it saves time scanning the road and seeing what is wrong and what I have to do". Between the static and moving lights, the moving light was preferred (71%), "I felt the movement of the light was instructing me to steer to the direction" one participant mentioned. The participants who preferred the static light mentioned that it was less distracting and annoying than the moving light. One of the participants said that "the moving light stresses me because it reminds me of police or ambulance alerts".

DISCUSSION

Ambient Light as TOR

The main objective of this work was to investigate ambient light cues as take over requests. Our motivation for using these displays, was the high demand on focal visual resource in driving context. Based on Wickens multiple resource theory, we designed an ambient display, which addresses the peripheral vision of drivers. We defined three light conditions one of which (baseline) basically was meant to shift the attention of the driver from a secondary task to the take-over situation; and the other two were designed for conveying contextual information about the traffic situation, in addition to attention shifting functionality.

In our experiment, we measured reaction time as a measure for perception and understanding of TORs, and TTC as a factor of driving behavior. Results showed that the conditions in

which displays convey contextual information about driving environment (static, moving), led to shorter reaction times and safer driving. These results confirm our first and third hypotheses (H1, H3). Based on this, we can conclude that the conveyed information can increase situation awareness and assist drivers in decision making and performing appropriate actions. The overall perceived workload was also rated less in these conditions than the baseline, however the difference was not statistically significant. This implies that, ambient displays could support drivers with shifting their attention from a secondary task and in the meanwhile convey information about the upcoming maneuver at take-over time, without increasing the workload. This is in consonance with our fifth hypothesis (H5).

We experimented two different light patterns for conveying information to the driver. We drew on related work and in addition to static light, designed a moving light cue. Results indicated that, unlike our assumption in hypotheses H2 and H4, the light pattern of the presented cue does not have an effect on reaction times and TTC. The ratings of perceived workload however, show that despite creating more frustration, the moving pattern is less demanding. The overall ratings of perceived workload between these two conditions nonetheless, did not have a significant difference which refutes our sixth hypothesis (H6).

Ambient Displays and Peripheral Vision

We used ambient displays, to address the peripheral vision of drivers to convey information about their upcoming take-over situation without overloading their focal resources.

Using the labels provided by [22], we have 'substantial' evidence that static cues attract more glances than the baseline but only 'anecdotal' evidence for moving cues. Furthermore, we have 'anecdotal' evidence that moving cues result in glances that have similar duration lengths as our baseline cues, and 'anecdotal' evidence that static cues result in longer glances. One reason for this suggestive evidence could be small effect size. Future research should focus on these plausible differences and be specifically designed to study how the strength of static or moving cues could vary in terms of the glances that they attract away from the outside world. Unfortunately, this current study was not explicitly designed for this purpose.

These results reveal that using moving peripheral cues as TORs are optimal; they prime drivers with contextual information and result in fewer glances for shorter duration in comparison with the static condition. Despite the fact that the baseline cue led to the least number and duration of glances, given the results from driving behavior (reaction times and TTC), it does not surpass the moving cue in terms of effect on performance.

Conveying Information at Take-over Situations

In this work, we visualized the steering direction at take-over situation by using ambient light, given the location of the road block. Our first idea was to shift drivers' attention to where the road block is (what to avoid). However, during our pilot study, we observed that, users perceived the displayed direction, as the direction to steer to (what to do) despite having a red color on the display-which is an indicator of danger or avoidance.

This led us to the conclusion that, users expect the system to direct them to what to do in hazardous situations or decision making bottlenecks. Therefore, we changed our set up to have the light to direct users to where to steer to. The question of "What information should be communicated to the user at take-over situation to support a smooth transition?" is however, not in the scope of this work but requires further research.

LIMITATIONS

As in early in-vehicle prototype evaluations, our study was run in a driving simulator under a specific road scenario. Real driving conditions however, can be more dynamic and complex than what was tested in our work. For example, the traffic, weather, and road conditions, or the type, duration and level of engagement in secondary task can be factors that can influence take-over behavior. However, the goal of this work was to provide initial validation for whether peripheral light display cues can be used as take over requests, and we would require further testing of our approach before deployment in a real test scenario.

CONCLUSION

Cues for TOR perform their role in two stages. First, attention disengagement from the non-driving task is required. This is typically achieved with salient sensory cues. Second, information can be provided to cue the appropriate action that the driver has to perform. In the current study, we employ three types of cues. A baseline cue that only promotes attentional disengagement from the non-driving task towards the visual world, and two cues (static vs. moving) that provides information about the action that should be taken. A static cue is expected attract overt attention to itself while a moving cue is expected to attract overt attention to the operational context. This is because the former attracts focal resources while the latter is processed with ambient resources (Wickens, 2002). We argue that although improved performance is equivalent with the use of both cues in terms of response time, ambient cues are more suitable. They cue focal attention appropriately to the operational context without capturing focal attention itself.

FUTURE WORK

This study sought to evaluate the effectiveness of ambient light cues as take over requests. With the collected results we showed that, ambient cues which convey contextual information can result in shorter reaction times and safer maneuvers without requiring more workload, and more or longer duration of glances. However, we only tested the light display which indicated one information item (steering direction) at take-over situation. Follow up work should investigate whether more information about the driving context can be encoded in ambient light displays to support take-over maneuvers.

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REFERENCES

1. National Highway Traffic Safety Administration and others. 2013. Preliminary statement of policy concerning automated vehicles. *Washington, DC* (2013).
2. Carryl L Baldwin and Bridget A Lewis. 2014. Perceived urgency mapping across modalities within a driving context. *Applied ergonomics* 45, 5 (2014), 1270–1277.
3. Carryl L Baldwin and Jennifer F May. 2011. Loudness interacts with semantics in auditory warnings to impact rear-end collisions. *Transportation research part F: traffic psychology and behaviour* 14, 1 (2011), 36–42.
4. James C Byers, AC Bittner, and SG Hill. 1989. Traditional and raw task load index (TLX) correlations: Are paired comparisons necessary. *Advances in industrial ergonomics and safety I* (1989), 481–485.
5. Daniel Damböck, Thomas Weißgerber, Martin Kienle, and Klaus Bengler. 2012. Evaluation of a contact analog head-up display for highly automated driving. In *4th International Conference on Applied Human Factors and Ergonomics. San Francisco. USA*.
6. Christian Gold, Daniel Damböck, Lutz Lorenz, and Klaus Bengler. 2013. "Take over!" How long does it take to get the driver back into the loop?. In *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, Vol. 57. SAGE Publications, 1938–1942.
7. PA Hancock, Lisa Simmons, L Hashemi, H Howarth, and T Ranney. 1999. The effects of in-vehicle distraction on driver response during a crucial driving maneuver. *Transportation Human Factors* 1, 4 (1999), 295–309.
8. Robert E Kass and Adrian E Raftery. 1995. Bayes factors. *Journal of the american statistical association* 90, 430 (1995), 773–795.
9. Jeamin Koo, Jungsuk Kwac, Wendy Ju, Martin Steinert, Larry Leifer, and Clifford Nass. 2015. Why did my car just do that? Explaining semi-autonomous driving actions to improve driver understanding, trust, and performance. *International Journal on Interactive Design and Manufacturing (IJIDeM)* 9, 4 (2015), 269–275.
10. Sabine Langlois. 2013. ADAS HMI Using Peripheral Vision. In *Proceedings of the 5th International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutomotiveUI '13)*. ACM, New York, NY, USA, 74–81.
11. Florian Laquai, Fabian Chowanetz, and Gerhard Rigoll. 2011. A large-scale LED array to support anticipatory driving. In *Systems, Man, and Cybernetics (SMC), 2011 IEEE International Conference on*. IEEE, 2087–2092.
12. HW Leibowitz, RB Post, Th Brandt, and J Dichgans. 1982. Implications of recent developments in dynamic spatial orientation and visual resolution for vehicle guidance. In *Tutorials on motion perception*. Springer, 231–260.

13. Andreas Löcken, Heiko Müller, Wilko Heuten, and Susanne Boll. 2015. An experiment on ambient light patterns to support lane change decisions. In *Intelligent Vehicles Symposium (IV)*. IEEE, 505–510.
14. Tara Matthews, Anind K. Dey, Jennifer Mankoff, Scott Carter, and Tye Rattenbury. 2004. A Toolkit for Managing User Attention in Peripheral Displays. In *Proceedings of the 17th Annual ACM Symposium on User Interface Software and Technology (UIST '04)*. ACM, New York, NY, USA, 247–256.
15. Natasha Merat, A Hamish Jamson, Frank CH Lai, Michael Daly, and Oliver MJ Carsten. 2014. Transition to manual: Driver behaviour when resuming control from a highly automated vehicle. *Transportation research part F: traffic psychology and behaviour* 27 (2014), 274–282.
16. Natasha Merat and John D Lee. 2012. Preface to the special section on human factors and automation in vehicles designing highly automated vehicles with the driver in mind. *Human Factors: The Journal of the Human Factors and Ergonomics Society* 54, 5 (2012), 681–686.
17. Alexander Meschtscherjakov, Christine Döttlinger, Christina Rödel, and Manfred Tscheligi. 2015. ChaseLight: Ambient LED Stripes to Control Driving Speed. In *Proceedings of the 7th International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutomotiveUI '15)*. ACM, New York, NY, USA, 212–219.
18. Frederik Naujoks, Christoph Mai, and Alexandra Neukum. 2014. The effect of urgency of take-over requests during highly automated driving under distraction conditions. *Advances in Human Aspects of Transportation Part I* (2014), 431.
19. Helen J. Neville and Donald Lawson. 1987. Attention to central and peripheral visual space in a movement detection task: an event-related potential and behavioral study. I. Normal hearing adults. *Brain Research* 405, 2 (1987), 253 – 267.
20. Ioannis Politis, Stephen Brewster, and Frank Pollick. 2015. Language-based multimodal displays for the handover of control in autonomous cars. In *Proceedings of the 7th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*. ACM, 3–10.
21. Ioannis Politis, Stephen A Brewster, and Frank Pollick. 2014. Evaluating multimodal driver displays under varying situational urgency. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. ACM, 4067–4076.
22. Eric-Jan Wagenmakers, Ruud Wetzels, Denny Borsboom, and Han LJ van der Maas. 2011. Why psychologists must change the way they analyze their data: the case of psi: comment on Bem. (2011), 426.
23. Mark Weiser and John Seely Brown. 1996. Designing calm technology. *PowerGrid Journal* 1, 1 (1996), 75–85.
24. Christopher D Wickens. 2002. Multiple resources and performance prediction. *Theoretical issues in ergonomics science* 3, 2 (2002), 159–177.